# Lecture 01 – Introduction to Machine Learning

January 17, 2022

# + Activity - 01

**Machines imitating and adapting human like behaviour.**

**How exactly?**

- For an example, let me ask you a quiz…

  3 – 9

  4 – 16

  8 – 64

  9 – ?

- How did you come to ___________

# + Topics to be covered

1. What is Learning?
2. Introduction to machine Learning
3. Machine Learning Vs. Traditional Programming
4. Why Machine Learning
5. How Machine Learning Works
6. Role & Responsibility of Machine Learning
7. Growth of Machine Learning
8. Relevant Disciplines
9. Applications of Machine Learning
10. Some Issues in Machine Learning
11. Types of Machine Learning
12. Machine Learning Process

# + A Few Quotes

التطور

- "A breakthrough in machine learning would be worth ten Microsoft" (Bill Gates, Chairman, Microsoft)
- "Machine learning is the next Internet" (Tony Tether, Director, DARPA)
- Machine learning is the hot new thing" (John Hennessy, President, Stanford)

ترتيبات الويب

- "Web rankings today are mostly a matter of machine learning" (Prabhakar Raghavan, Dir. Research, Yahoo)

ثورة حقيقية

- "Machine learning is going to result in a real revolution" (Greg Papadopoulos, CTO, Sun)

فجوة

- "Machine learning is today's discontinuity" (Jerry Yang, CEO, Yahoo)

# + What is Learning?

# + Learning

**Learning is a verb**

- the **activity** of obtaining knowledge
- **knowledge** obtained by study

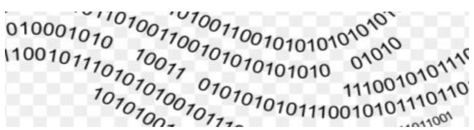
## Learning in Human

**think**

**see**

**experience**

## Learning in Computer



# + What is Learning



- "The acquisition of knowledge or skills through study, experience, or being taught." - google
- "Learning is making useful changes in our minds", - Marvin Minsky
- A computer program is said to learn from ***experience E*** with respect to some class of ***tasks T*** and ***performance measure P***, if its performance at tasks in ***T***, as measured by ***P***, improves with ***experience E***.
- 'The ***action*** of ***receiving*** instruction or ***acquiring*** knowledge'.
- 'A process which leads to the ***modification*** of ***behaviour*** or the ***acquisition*** of ***new abilities*** or ***responses***, and which is additional to natural development by ***growth*** or ***maturation***'.

النطور والنمو

# + Examples

- **A soccer learning problem:**

تعلم كرة القدم

  - **Task T**: Playing soccer
  - **Performance measure P**: Percent of games won against opponents
  - **Training experience E**: Playing practice games against itself

- **A handwriting recognition learning problem:**

التعرف على خط اليد فى الصور

  - **Task T**: Recognizing and classifying handwritten words within images
  - **Performance measure P**: Percent of words correctly classified
  - **Training experience E**: A database of handwritten words with given classifications

# + Examples

مشكلة تعلم القيادة للروبوت

- ***A robot driving learning problem:***

  اربعة مسارات

  - ***Task T***: Driving on public four-lane highways using vision sensors
  - ***Performance measure P***: Average distance travelled before an error (as judged by human

    مشرف بشرى

    overseer)

    سلسلة من الصورة واوامر القيادة المسجلة

  - ***Training experience E***: A sequence of images and steering commands recorded while observing a human driver

# + Examples

- Learning to recognize spoken words.
- Learning to drive an autonomous vehicle
- Learning to classify new astronomical structures
- Learning to play world-class backgammon.
- Learning to play chess game.

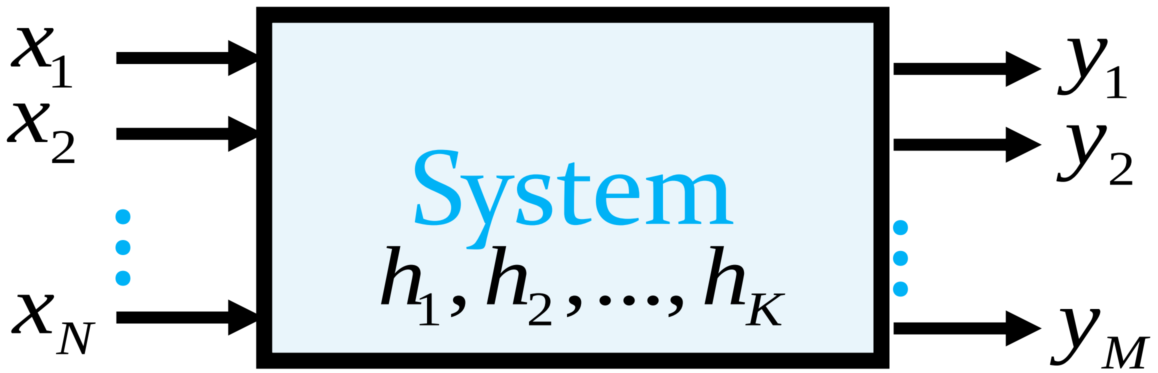
تعلم تصنيف الهياكل الفلكية الجديدة كالنجوم وغيره
تعلم للعب الطاولة الخلفية على مستوى عالمي

# + Learning System

- In order to complete the design of the learning system, we must now choose
  1. the exact type of knowledge to be learned
  2. a representation for this target knowledge
  3. a learning mechanism

- **Checkers Game:** لعبة الشطرنج
  - T: Play soccer
  - P: Percent of games won in world tournament
- What experience?
- What exactly should be learned?
- How shall it be represented?
- What specific algorithm to learn it?

# + Generic Learning System




Input Variables: $$\mathbf{x} = \left(x_1, x_2, \dots, x_N\right)$$

Hidden Variables: $$\mathbf{h} = \left(h_1, h_2, \dots, h_K\right)$$

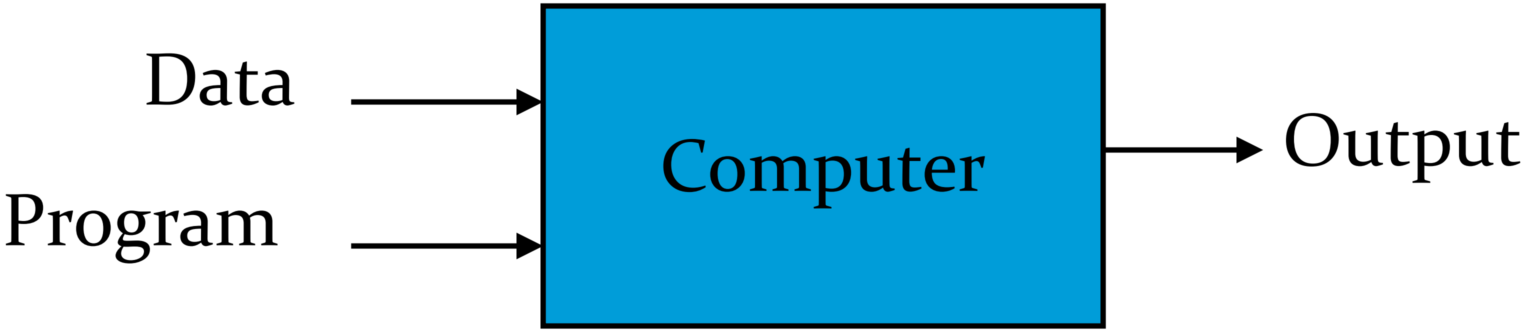
Output Variables: $$\mathbf{y} = \left(y_1, y_2, \dots, y_K\right)$$

# + What is Machine Learning?

# + Preamble: Machine Learning is what

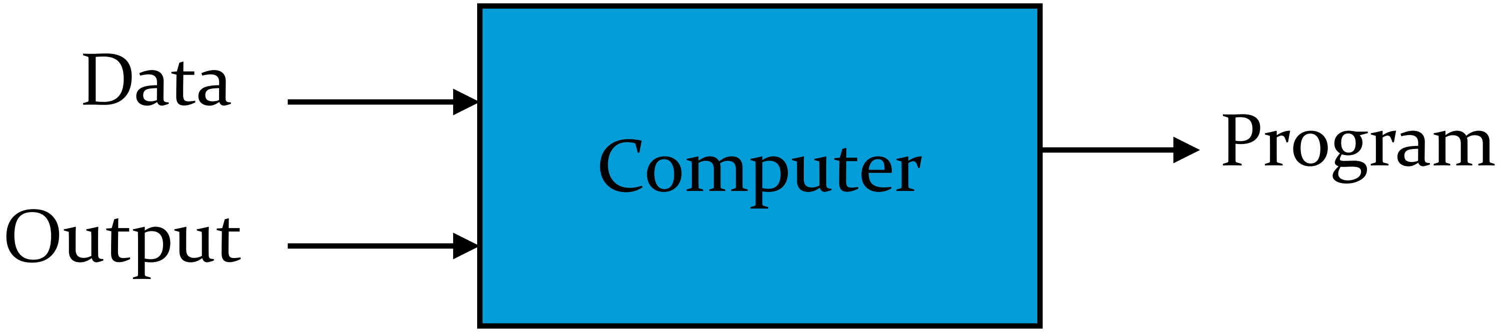
- Automating automation
- Getting computers to program themselves
- Writing software is the bottleneck
- Let the data do the work instead!

# Traditional Programming




1 + 2 = 3 ,  1,2 data, + operand

# Machine Learning




1  = 3
2  = 4
3  = 5

y = x +2
x=7
y+ 7+2 = 9

# + Preamble: Magic?

**No, more like gardening**

- بذور **Seeds** = Algorithms
- مغذيات **Nutrients** = Data
- **Gardener** = You
- **Plants** = Programs

# + What is Machine Learning?

- *A set of **methods** for the **automated analysis** of **structure** in data. …. **two** main **strands** of work,*
  i. ***unsupervised** learning ….*
  ii. ***supervised** learning.*
- *….similar to … **data mining**, but … **focus ..***
- ***More on autonomous machine performance**, rather than enabling humans to learn from the data.* ذاتی`
  - [Dictionary of Image Processing & Computer Vision, Fisher et al., 2014]

# + Machine Learning

Data
- Textual
- Visual
- Audio
- Video

Representation
- Histogram
- Bag of Words

Feature selection
- Dimensionality Reduction

**Machine Learning**
**=**
**Decision**
**Or**
**Prediction**

Predicted Information

# + What is Machine Learning?



**Predictive data analytics** is the art of **building** and using **models** that make **predictions** based on **patterns extracted** from historical data.

## What is Machine Learning?

- Machine learning, a branch of ***artificial intelligence***, concerns the ***construction*** and ***study*** of systems that can ***learn*** from data.
- Machine learning is ***programming computers to optimize a performance*** criterion using ***example data*** or ***past experience***.
- Machine learning is defined as an ***automated process that extracts patterns from data***.

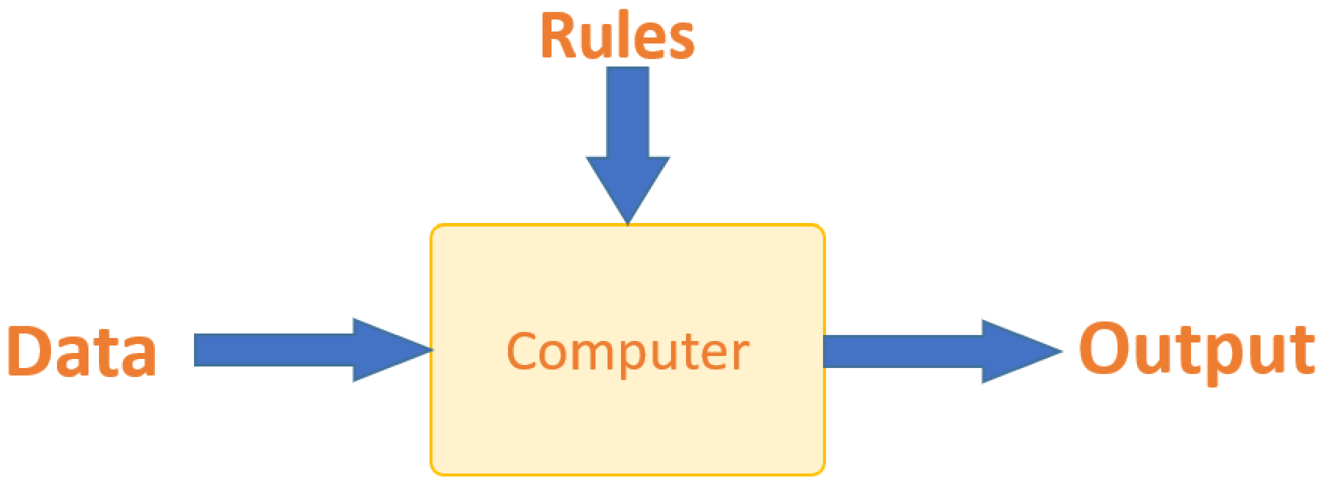
**Examples** ⇨ **Experience** ⇨ **Prediction**

# + Machine Learning vs. Traditional Programming

- Traditional programming differs significantly from machine learning, where ***programmer code all the rules for which software is being developed***.

- Each rule is based on a ***logical foundation***

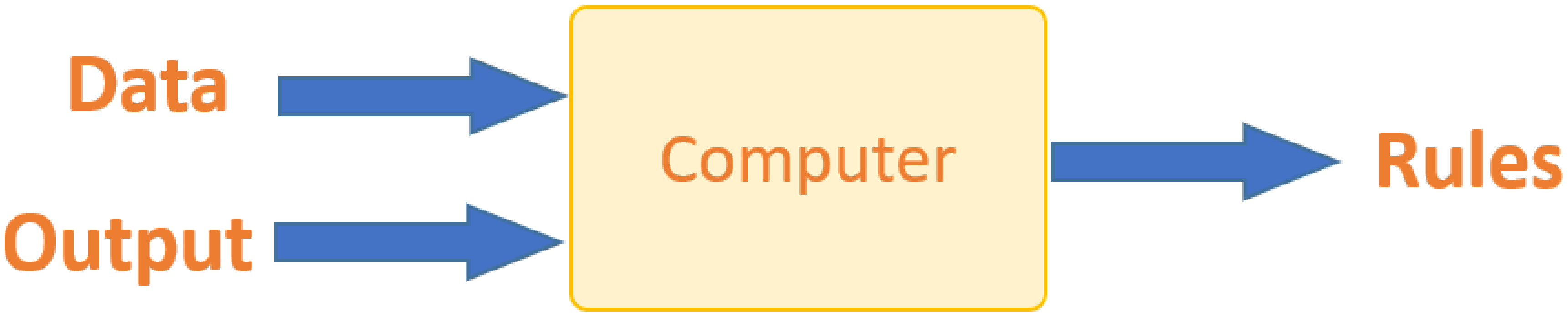
- When the system grows ***complex***, more rules need to be ***written***. It can quickly become ***unsustainable*** to ***maintain***.
غير مستقر




Traditional Programming

# + Machine Learning vs. Traditional Programming

- The ***goal*** of the Machine Learning is to ***build computer System*** that can ***adopt*** and ***learn*** from their ***experience*** - Tom Dietterich
- The machine learns ***how the input and output data are correlated and it writes a rule***.
- The ***programmers*** do ***not*** need to ***write*** new rules ***each time*** there is new data. The algorithms ***adapt in response to new data and experiences to improve efficacy over time***.




Machine Learning

# + Why “Machine Learning”? Or Why “Learn”?

- Machine learning is ***programming computers*** to ***optimize*** a performance ***criterion*** using example data or past experience.

- There is no need to “***learn***” to calculate payroll

- Learning is used when:
  - Human expertise does not exist (navigating on Mars),
  - Humans are unable to explain their expertise (speech recognition)
  - Solution changes in time (routing on a computer network)
  - Solution needs to be adapted to particular cases (user biometrics, user medical vital signs taken, etc.)

# + Why Machine Learning

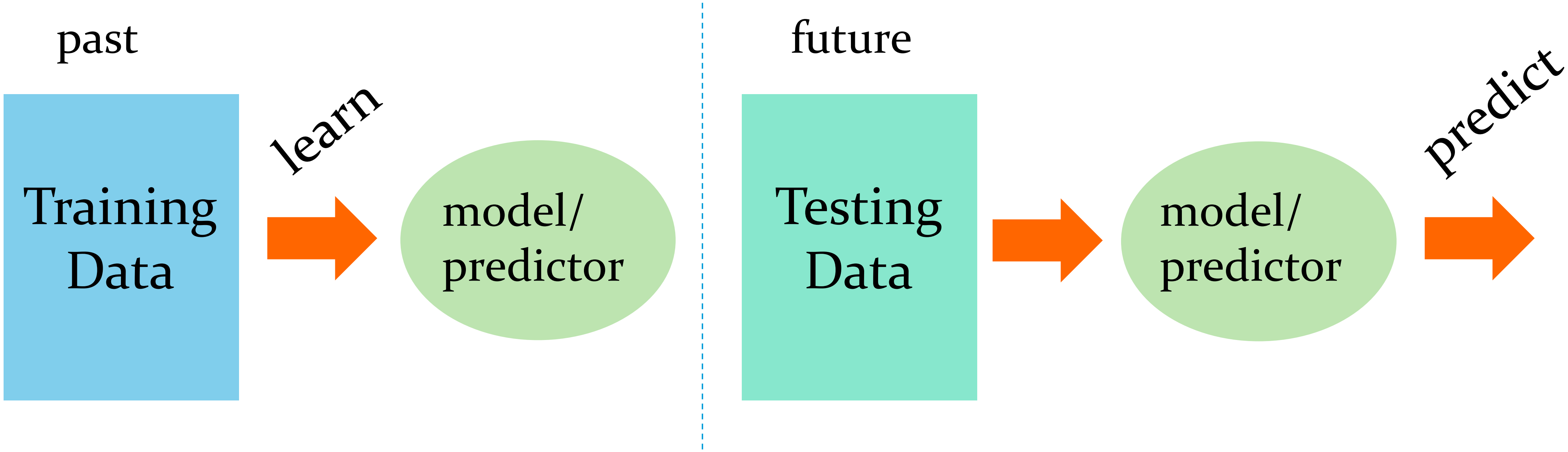
- ***Why Machine Learning*?**
  - we cannot program everything
  - some tasks are difficult to define algorithmically
- especially in computer vision
- **…. visual sensing has few rules**
- ***Sometimes look similar is not semantically similar***

  - Well-defined learning problems ?
  - – easy to learn Vs. difficult to learn
  - ….. varying complexity of visual patterns

# + How Machine Learning Works?

- The goal of machine learning is to ***develop methods*** that can ***automatically detect patterns*** in data, and then to ***use*** the ***uncovered patterns*** to ***predict*** future data or other outcomes of interest.-- Kevin P. Murphy

- Machine learning is about ***predicting*** the ***future based*** on the ***past***. Hal Daume III



# + How Machine Learning Works?

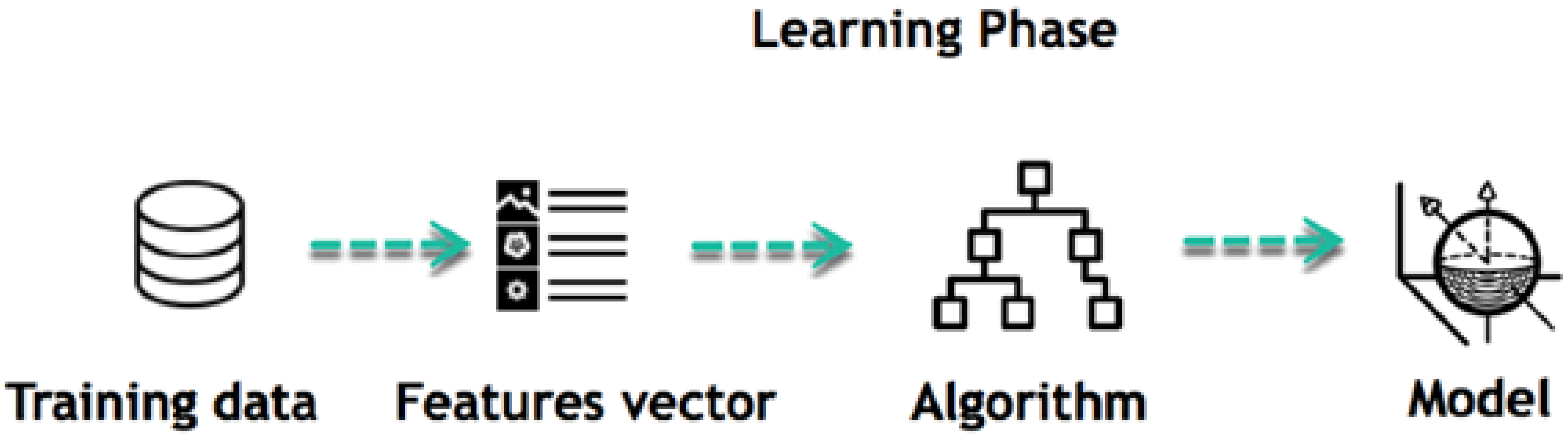
- ***The way the machine learns is similar to the human being.***
- Humans ***learn*** from ***experience***. The more we know, the more easily we can predict.
- To make an ***accurate prediction***, the machine sees an example.

The core objective of machine learning is the

- **Learning**
- **Inference**.

# + How Machine Learning Works?

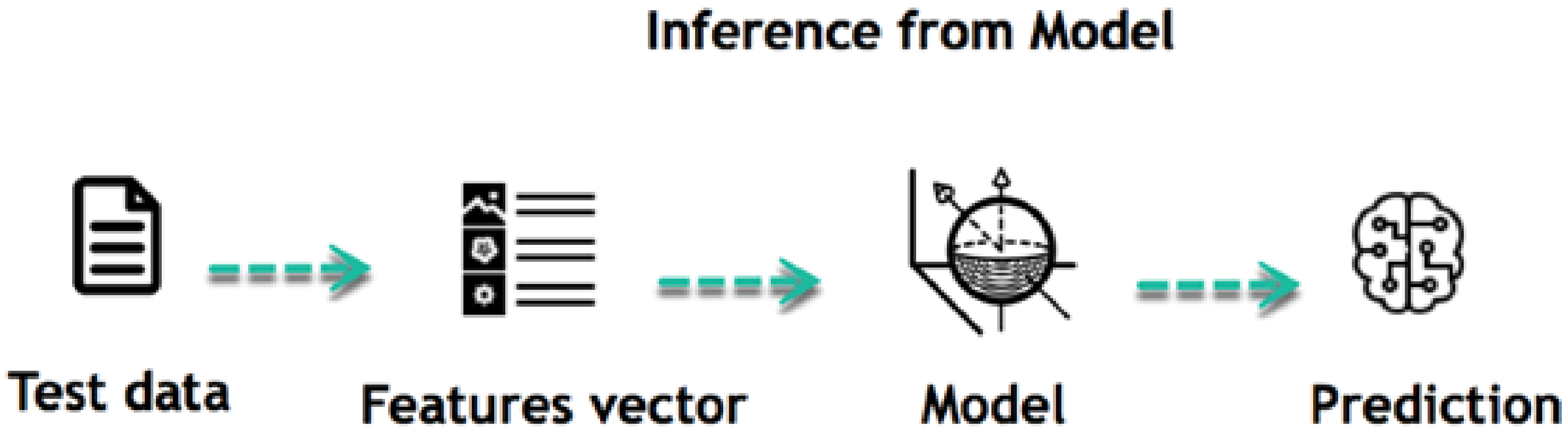
- **Learning Phase:** First of all, the machine learns through the ***discovery*** of ***patterns*** in the data. We need to choose ***carefully*** which ***data*** to provide to the ***machine***. The ***list of attributes used to solve a problem is called a feature vector***.



# + How Machine Learning Works?

- **Inference Phase:** When the ***model is built***, we need to ***test*** it on ***never-seen-before data***. The new data are ***transformed into a features vector***, go through the model and give a ***prediction***.

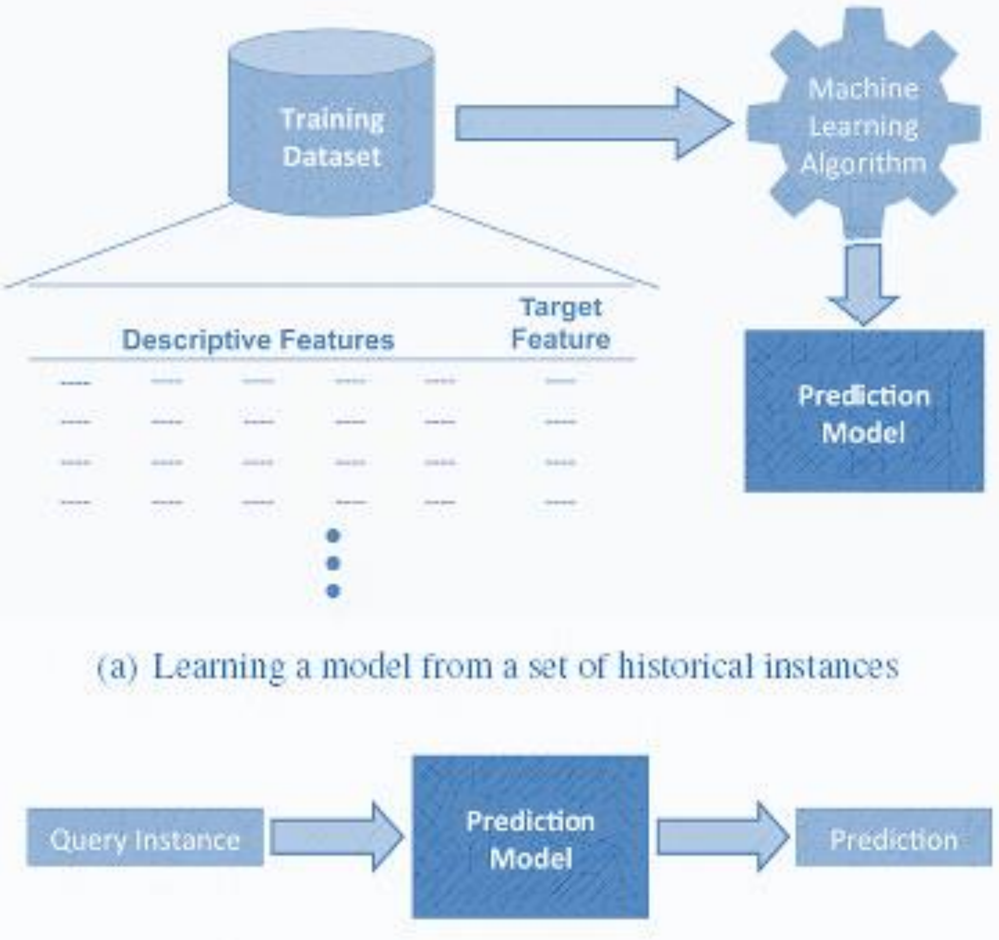
- ***There is no need to update the rules or train again the model.*** You can use the model ***previously trained*** to make ***inference*** on ***new*** data.



# + Review: Machine Learning

- Machine learning algorithm ***learn*** the ***relationship*** between a ***set of descriptive features*** and a ***target feature*** based on a set of ***historical examples***.

- We can then use ***this model to make prediction for new data*** (instance/record)




(a) Learning a model from a set of historical instances

(b) Using a model to make predictions

# + Activity - 02

التمويل العقارى

- **Data set of mortgages that a bank has granted in the past.**
  - Data set includes descriptive features and a target feature.
    - Descriptive features: tell us three pieces of information, i.e. occupation (professional or industrial), Age and ratio between the applicant salary and then amount of the loan taken
    - Outcome (target feature): is set to either default or repay.
- **In ML terms, each row in the data set is referred to as a training instance, and the overall dataset is referred to as a training data set.**

| ID | Occupation | Age | Loan-Sal-Ratio | Outcome |
|---|---|---|---|---|
| 1 | Industrial | 34 | 2.96 | repay |
| 2 | Professional | 41 | 4.64 | default |
| 3 | Professional | 36 | 3.22 | default |
| 4 | Professional | 41 | 3.11 | default |
| 5 | Industrial | 48 | 3.8 | default |
| 6 | Industrial | 61 | 2.52 | repay |
| 7 | Professional | 37 | 1.5 | repay |
| 8 | Professional | 40 | 1.93 | repay |
| 9 | Industrial | 33 | 5.25 | default |
| 10 | Industrial | 32 | 4.15 | default |

**A very simple prediction model for this domain would be**

**IF** (Loan-Sal-Ration > 3) **THEN**
Outcome = default
**ELSE** Outcome = repay

The model is **consistent** with the dataset as there's no instance for which the model failed. When new mortgage application come, we can use this model to predict whether the applicant will repay or will be default and make decision based on this prediction.

# + Review: Machine Learning

- Machine Learning ***automate*** the ***process of learning*** a model that ***captures*** the ***relationship*** between the ***descriptive features*** and the ***target feature*** in a dataset.
- For simple datasets, we may be able to ***manually create a prediction model*** and in an example of this scale, machine learning has little to offer us.
- But….
  - In case ***data size is large***, then …

# + Activity - 03

- A more complex dataset for the same problem with more descriptive features.
- The earlier (activity-02) prediction model is no longer consistent with this dataset.
- We need to build another one, like…

**IF** (Loan-Sal-Ration < 1.5) **THEN** Outcome = repay
**ELSE IF** (Loan-Sal-Ration > 4) **THEN** Outcome = default
**ELSE IF** (Age < 40 **AND** Occupation=Industrial) **THEN** Outcome = default
**ELSE** Outcome = repay

- To manually learn this model by examining the data is almost impossible.
- For a machine learning algorithm, this is simple.
- To build a prediction model for large dataset with multiple features, ML- is the solution

| ID | Amount | Salary | Loan-Sal-Ratio (Amt / Sal) | Age | Occupation | Property | Type | Outcome |
|---|---|---|---|---|---|---|---|---|
| 1 | 245,100 | 66,400 | 3.69 | 44 | Industrial | Farm | stb | repay |
| 2 | 90,600 | 75,300 | 1.20 | 41 | Industrial | Farm | stb | repay |
| 3 | 195,600 | 52,100 | 3.75 | 37 | Industrial | Farm | ftb | default |
| 4 | 157,800 | 67,600 | 2.33 | 44 | Industrial | Apartment | ftb | repay |
| 5 | 150,800 | 35,800 | 4.21 | 39 | Professional | Apartment | stb | default |
| 6 | 133,000 | 45,300 | 2.94 | 29 | Industrial | Farm | ftb | default |
| 7 | 193,100 | 73,200 | 2.64 | 38 | Professional | House | ftb | repay |
| 8 | 215,000 | 77,600 | 2.77 | 17 | Professional | Farm | ftb | repay |
| 9 | 83,000 | 62,500 | 1.33 | 30 | Professional | House | ftb | repay |
| 10 | 186,100 | 49,200 | 3.78 | 30 | Industrial | House | ftb | default |
| 11 | 161,500 | 53,300 | 3.03 | 28 | Professional | Apartment | stb | repay |
| 12 | 157,400 | 63,900 | 2.46 | 30 | Professional | Farm | stb | repay |
| 13 | 210,000 | 54,200 | 3.87 | 43 | Professional | Apartment | ftb | repay |
| 14 | 209,700 | 53,000 | 3.96 | 39 | Industrial | Farm | ftb | default |
| 15 | 143,200 | 65,300 | 2.19 | 32 | Industrial | Apartment | ftb | default |
| 16 | 203,000 | 64,400 | 3.15 | 44 | Industrial | Farm | ftb | repay |
| 17 | 247,800 | 63,800 | 3.88 | 46 | Industrial | House | stb | repay |
| 18 | 162,700 | 77,400 | 2.10 | 37 | Professional | House | ftb | repay |
| 19 | 123,300 | 61,100 | 2.02 | 21 | Industrial | Apartment | ftb | default |
| 20 | 284,100 | 32,300 | 8.80 | 51 | Industrial | Farm | ftb | default |
| 21 | 154,000 | 48,900 | 3.15 | 49 | Professional | House | stb | repay |
| 22 | 112,800 | 79,700 | 1.42 | 41 | Professional | House | ftb | repay |
| 23 | 252,000 | 59,700 | 4.22 | 27 | Professional | House | stb | default |
| 24 | 175,200 | 39,900 | 4.39 | 37 | Professional | Apartment | stb | default |
| 25 | 149,700 | 58,600 | 2.55 | 35 | Industrial | Farm | stb | default |

ftb = first-time buyer, stb = second-time buyer

# + Role & Responsibility of Machine Learning?

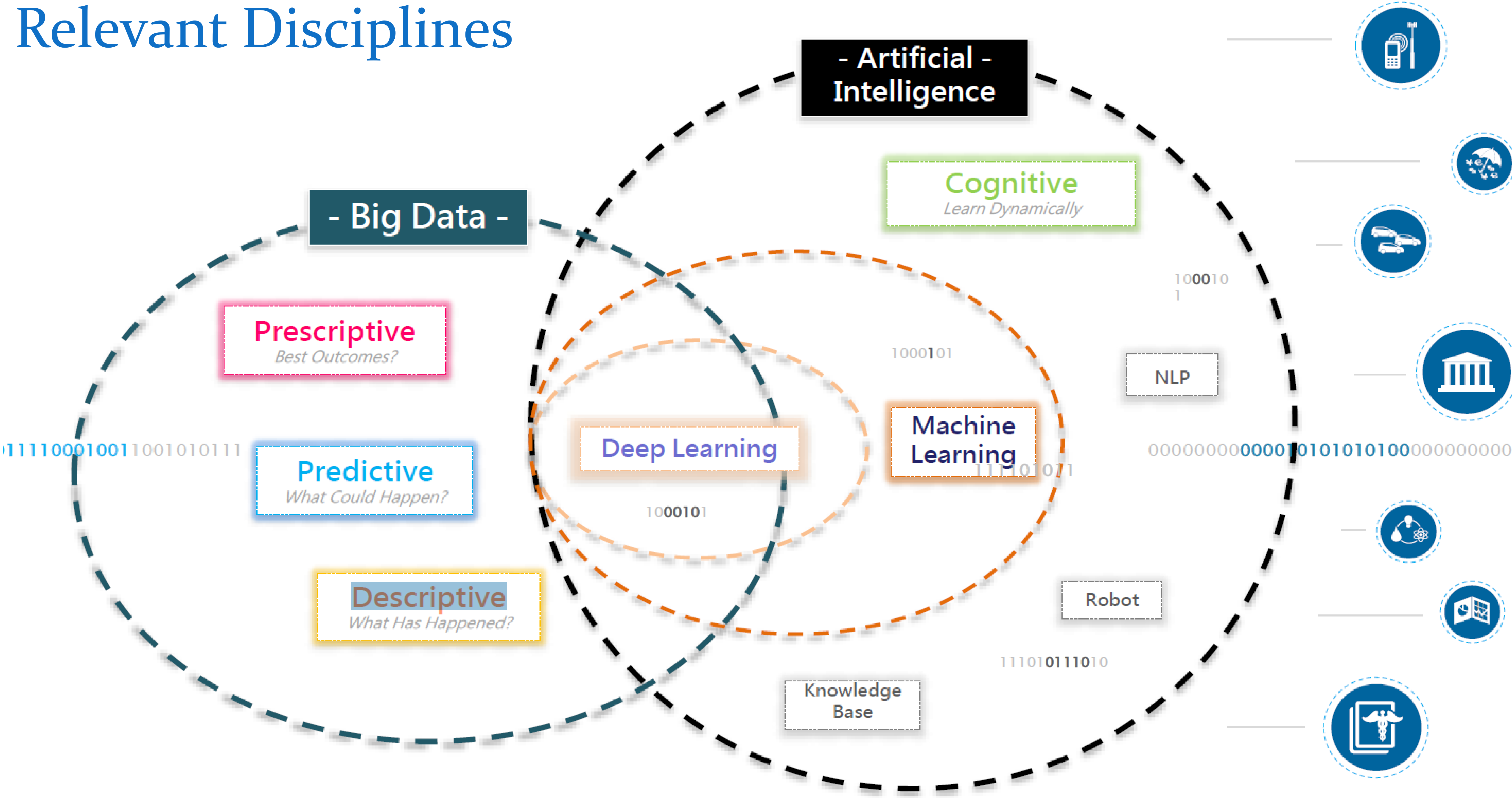


- Machine Learning
  - Study of algorithms that
  - improve their performance
  - at some task
  - with experience
- Optimize a ***performance criterion*** using example data or past experience.
- Role of Statistics: ***Inference from a sample***
- Role of Computer science: ***Efficient algorithms to***
  - Solve the optimization problem
  - Representing and evaluating the model for inference

# + Growth of Machine Learning

- Machine learning is preferred approach to
  - Speech recognition, Natural language processing
  - Computer vision
  - Medical outcomes analysis
  - Robot control
  - Computational biology
  - Text Mining

# Relevant Disciplines



# + Applications of Machine Learning

1. Speech and hand-writing recognition
2. Telephone menu navigation
3. Computer vision
4. Mail sorting
5. Bio-surveillance
6. Identifying disease outbreaks
7. Robot control
8. Autonomous driving
9. Data mining
10. Bioinformatics
11. Playing games
12. Fault detection
13. Clinical diagnosis
14. Spam email detection
15. Retail: Market basket analysis, Customer relationship management (CRM)
16. Finance: Credit scoring, fraud detection
17. Manufacturing: Optimization, troubleshooting
18. Medicine: Medical diagnosis
19. Telecommunications: Quality of service optimization
20. Web mining: Search engines

# + Some Issues in Machine Learning

- What algorithms can approximate functions well and when?
- How does number of training example influence accuracy?
- How does complexity of hypothesis representation impact it?
- How does noisy data influence accuracy?
- What are the theoretical limits of learnability?
- How can prior knowledge of learner help?
- What clues can we get from biological learning systems?
- How can systems alter their own representations?

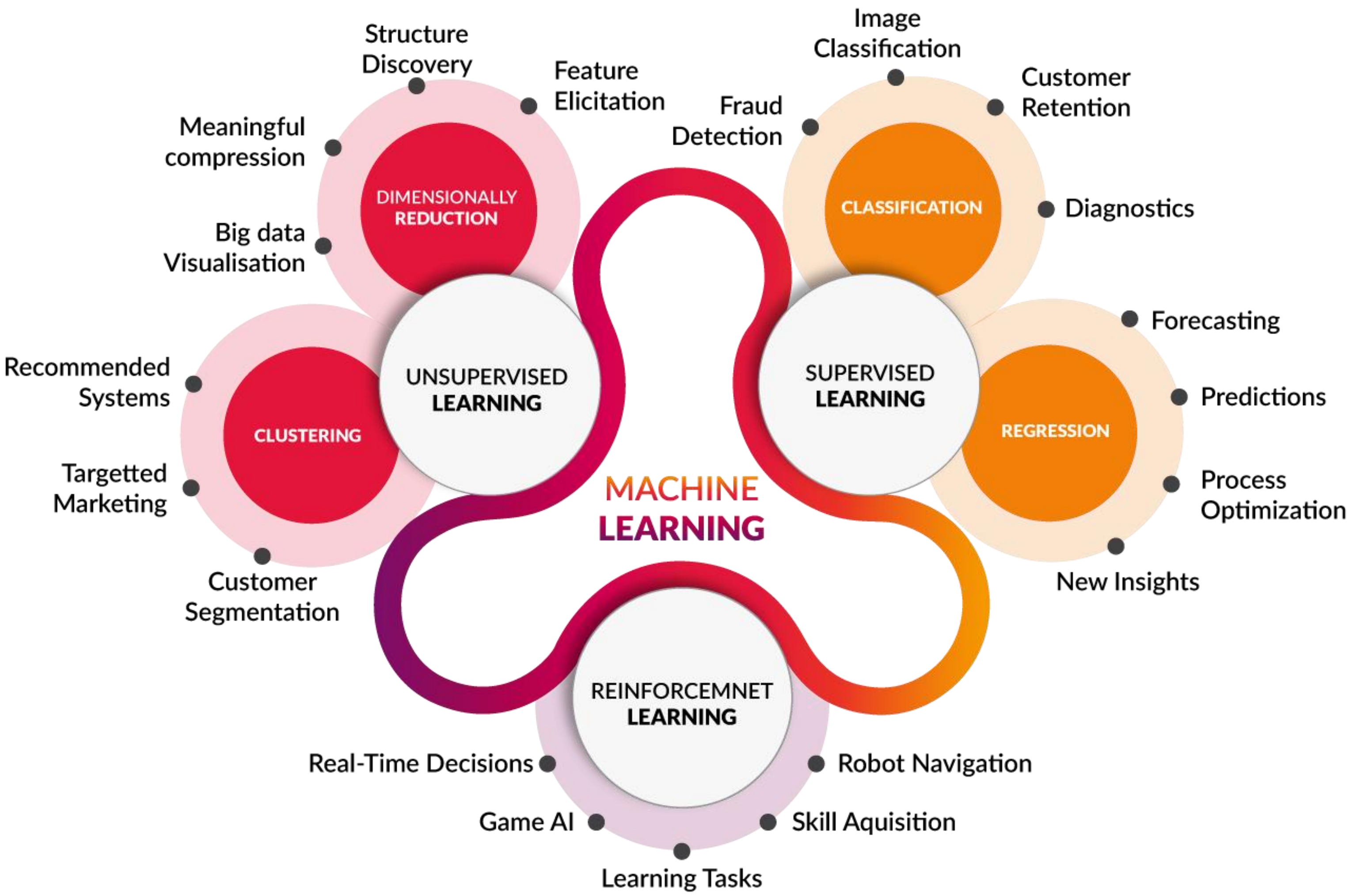
| | | |
|---|---|---|
| **Types of Machine Learning** | **Supervised Learning** (Inferential –Task Driven) | Regression |
| | | Classification |
| | **Unsupervised Learning** (Descriptive –Data Driven) | Clustering |
| | | Dimensionality Reduction |
| | | Anomaly Detection |
| | **Semi-supervised Learning** | Co-Training |
| | | Active Learning |
| | **Reinforcement Learning** (Learns from mistakes-Algorithm learns to react to environment) | Markov Decision Process |

# + Machine Learning Paradigm

| | Unsupervised | Supervised |
|---|---|---|
| **Continuous** | Clustering Dimensionality Reduction, Association Analysis | Regression |
| **Categorical** | Clustering Dimensionality Reduction, Association Analysis | Classification |

# + Types of Machine Learning

Main task performs in each types of Machine Learning



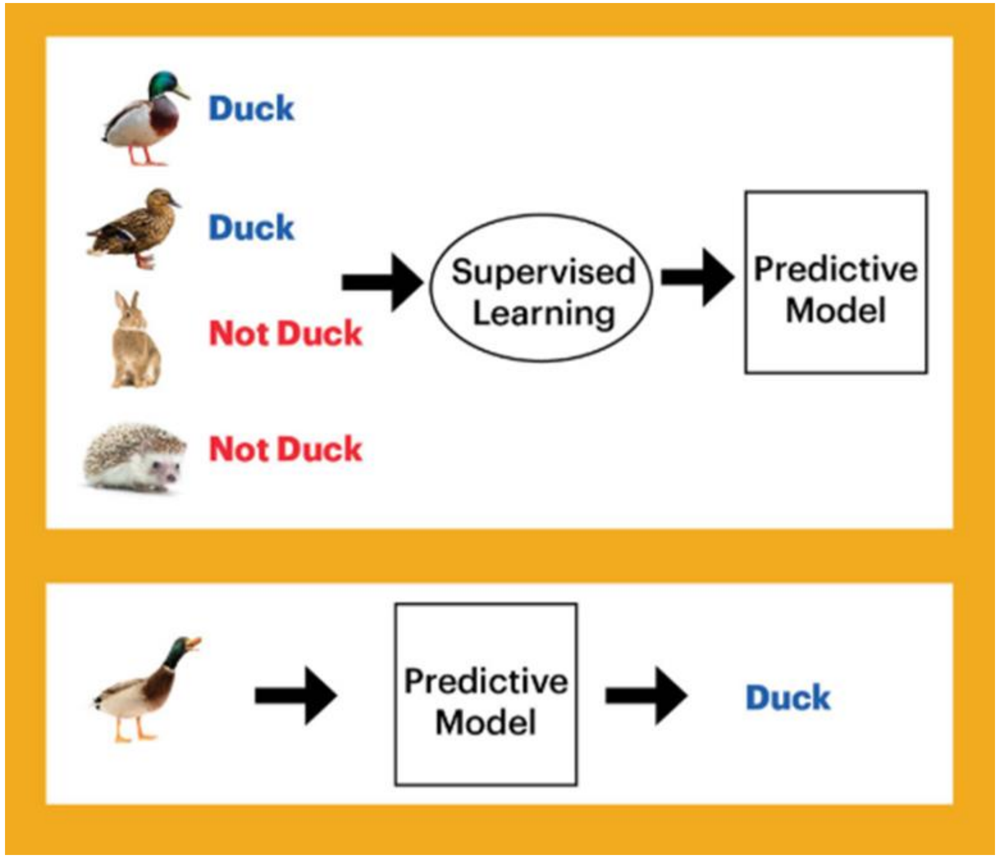
# Types of Machine Learning

- **Supervised learning:** (predictive model, "labeled" data)
  - classification (Logistic Regression, Decision Tree, KNN, Random Forest, SVM, Naive Bayes, etc.)
  - numeric prediction (Linear Regression, KNN, Gradient Boosting & AdaBoost, etc.)
- **Unsupervised learning:** (descriptive model, "unlabeled" data)
  - clustering (K-Means)
  - pattern discovery
- **Semi-supervised learning:** (mixture of "labeled" and "unlabeled" data).
- **Reinforcement learning:** Using this algorithm, the machine is trained to make specific decisions.
  - The machine is exposed to an environment where it trains itself continually using trial and error.
  - This machine learns from past experience and tries to capture the best possible knowledge to make accurate decisions.
  - Example of Reinforcement Learning: Markov Decision Process.

# + **Supervised Learning** *(Predictive Modeling)*

- **Supervised Learning** is the first type of machine learning, in which ***labelled*** data used to train the algorithms.

- The algorithms are trained using marked data, where the input and the output are known.
  - The input set of data is called as **Features** (denoted by X) along with the corresponding **outputs(target/Class labels)**(indicated by Y)
  - The algorithm learns by comparing its actual production with correct outputs to find errors.
- The raw data divided into two parts.
  - The first part is for **training** the algorithm, and ***(70%)***
  - The second is used for **test** the trained algorithm ***(30%)***



# + Supervised Learning



## Classification

- Build predictive models from training data which have ***features*** and ***class labels.***
- Use the features learnt from ***training data*** on new, previously ***unseen*** data to ***predict*** their class labels.
- The ***output classes*** are ***discrete***.
- Types of classification algorithms include decision trees, random forests, support vector machines, and many more.

## Regression

- used to predict output values based on some input features obtained from the data.
- Algorithm builds a model based on ***features*** and ***output*** values of the training data and this model is used to ***predict values for new data***.
- Output values in this case are ***continuous and not discrete***.
- Types of regression algorithms include linear regression, multivariate regression, regression trees, and lasso regression, among many others.

# + Unsupervised Learning *(Descriptive Modeling)*

- **Clustering:**
  - The goal here is to divide the input dataset into ***logical groups*** of ***related items***.
  - Some examples are grouping ***similar news articles***, grouping similar customers based on their profile, etc.

- **Dimension Reduction:**
  - Here the goal is to ***simplify*** a ***large input dataset*** by mapping them to a ***lower dimensional space***.
  - For example, carrying analysis on a large dimension dataset is very computationally intensive, so to simplify you may want to find the key variables that hold a significant percentage (say 95%) of information and only use them for analysis.

- **Anomaly Detection:**
  - Anomaly detection is also commonly known as ***outlier detection*** is the identification of items, events or observations which ***do not conform*** to an ***expected pattern*** or behavior in comparison with other items in a given dataset.
  - It has applicability in a variety of domains, such as ***machine or system health monitoring***, event detection, fraud/intrusion detection etc.

# + Semi-supervised Learning

- Semi-supervised machine learning is a ***combination of supervised and unsupervised*** machine learning methods.
- In semi-supervised learning, an algorithm learns from a dataset that includes both ***labeled and unlabeled data***, usually mostly unlabeled.

## Why is Semi-Supervised Machine Learning important?

- When you don't have enough labeled data to produce an accurate model and you don't have the ability or resources to get more, you can use semi-supervised techniques to increase the size of your training data.

| Name | Loan Amount | Loan Repaid | Fraud |
|---|---|---|---|
| Ashley | 100000 | 1 | 1 |
| Chuck | 25000 | 0 | 0 |
| Tim | 4000 | 1 | 1 |
| Mike | 150000 | 1 | 1 |
| Colin | 200000000 | 0 | |
| Libby | 400400 | 1 | 0 |
| Sheila | 3200 | 1 | 1 |
| Mandi | 34850 | 1 | |
| Gareth | 6570 | 0 | 0 |

- You can use a semi-supervised learning algorithm to label the data, and retrain the model with the newly labeled dataset

| Name | Loan Amount | Loan Repaid | Fraud |
|---|---|---|---|
| Ashley | 100000 | 1 | 1 |
| Chuck | 25000 | 0 | 0 |
| Tim | 4000 | 1 | 1 |
| Mike | 150000 | 1 | 1 |
| Colin | 200000000 | 0 | 0 |
| Libby | 400400 | 1 | 0 |
| Sheila | 3200 | 1 | 1 |
| Mandi | 34850 | 1 | 1 |
| Gareth | 6570 | 0 | 0 |

There is no way to verify that the algorithm produced labels that are 100% accurate, resulting in less trustworthy outcomes than traditional supervised techniques.

*https://www.datarobot.com/wiki/semi-supervised-machine-learning/*

# Reinforcement Learning

- ***Observations*** gathered from the ***interaction*** with the environment to take ***actions*** that would ***maximize the reward*** or ***minimize*** the risk.

- Reinforcement learning algorithm (***called the agent***) ***continuously learns*** from the ***environment*** in an ***iterative fashion***.

- In the process, the ***agent learns*** from its ***experiences*** of the ***environment*** until it explores the ***full range of possible states***.

- In order to produce ***intelligent programs*** (also called ***agents***), reinforcement learning goes through the following steps:
  1. Input state is observed by the agent.
  2. Decision making function is used to make the agent perform an action.
  3. After the action is performed, the agent receives reward or reinforcement from the environment.
  4. The state-action pair information about the reward is stored.

- Some applications of the *reinforcement learning algorithms* are computer played board games (Chess, Go), robotic hands, and self-driving cars.

- For example, a program to play a game or drive a car will have to constantly interact with a ***dynamic environment*** in which it is ***expected*** to perform a ***certain goal***.

# + Machine Learning Process

1. ***Collecting Data***: Data-set having variety, density and volume of relevant data will help in better learning.

2. ***Preparing the data:*** This involves fixing issues with the data set collected e.g. handling outliers and managing missing data points. Break the cleaned data-set into two parts, one for training and other for evaluating the program. Visualize the data.

3. ***Training a model:*** Choose an appropriate algorithm and representation of data in form of the model suited for your problem. Use the training data-set to train the model.

4. ***Evaluating the model:*** To test the accuracy and precision of the model, use the test data-set kept aside in the step 2.

5. ***Improving the performance:*** It might involve choosing different model and algorithm altogether, or introducing more variables and/or data to train the model.

# + Machine Learning Process

**Data**

## Data Exploration

1. Explore the Data
2. Visualize the Data
3. Feature Selection
4. Feature Extraction

## Learning

- Supervised Learning
- Un-supervised Learning
- Semi Supervised Learning
- Reinforcement Learning

## Evaluation

- Precision/Recall …
- Overfitting
- Test validation Data

**Model**

# Thanks